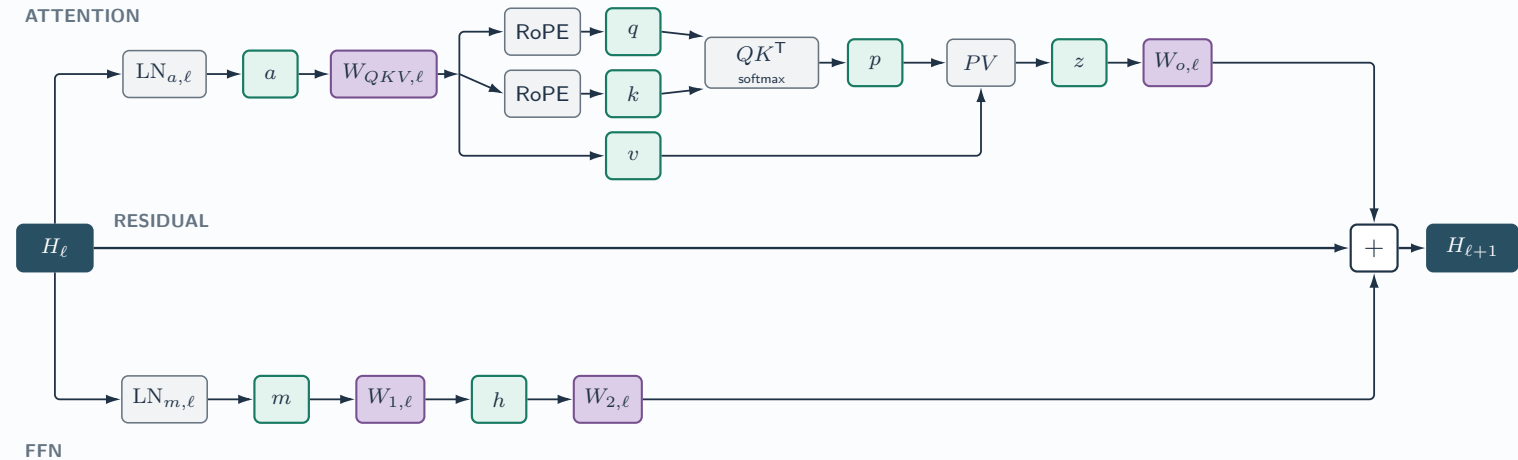


Shared Pythia block

● activation site ● weight ● operator



Sparsification ladder

Non-linearity ● GeLU ● ReLU ● $G_{+,\kappa}$ ● $G_{\pm,\kappa}$ ● none

Labels		Intervention								$R_{\text{model}}^{\text{max}}$ at $T = 2,048$		
Step	ID	h	m	a	z	q	k	v	Pressure	14M	70M	410M
1	A0	●	●	●	●	●	●	●	none	0.0%	0.0%	0.0%
2	A1-H	●	●	●	●	●	●	●	none	4.3%	12.4%	24.9%
3	A1-H-L1	●	●	●	●	●	●	●	L1N	4.3%	12.4%	24.9%
4	A1-H-OL1	●	●	●	●	●	●	●	OL1	4.3%	12.4%	24.9%
5	A4	●	●	●	●	●	●	●	none	12.8%	37.1%	74.8%
6	A4-OL1	●	●	●	●	●	●	●	OL1	12.8%	37.1%	74.8%
7	A7	●	●	●	●	●	●	●	none	30.0%	49.4%	87.2%
8	A7-OL1	●	●	●	●	●	●	●	OL1	30.0%	49.4%	87.2%